

FUNCTIONAL & TECHNICAL SPECIFICATION

Cloud-Based Umrah & Hajj Travel ERP

System specification and process flows for a secure, AI-assisted cloud platform on MySQL — with a dynamic field engine and dynamic bilingual (English / Arabic) printing.

System for

AL RAWDA GROUP HAJJ & UMRAH SERVICES

Bank Street, Street No. 910, Doha — Qatar

Document Information

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1. Introduction & Purpose

This document specifies the cloud-based Umrah & Hajj travel ERP that replaces the current Windows desktop application ("Stimes ERP Solution"). It defines what the system must do (functional specification), how it is structured (technical specification), and how work moves through it (process flows). It is the reference for design, build and testing.

1.1 Scope of the system

The system runs the complete Umrah/Hajj sales-to-collection cycle: maintaining master data, issuing invoices, collecting payments through receipts, recording expenses and refunds, approving transactions, and producing bilingual vouchers and management reports. It is delivered as a secure cloud application on a MySQL database, accessible from desktop, tablet and mobile, with a dynamic field engine, a dynamic print engine, and AI-assisted automation.

1.2 How requirements are written

Functional requirements use the identifier **FR-`<module>-<n>`** ; non-functional use **NFR-`<n>`** . Priority is shown as **M** Must, **S** Should, **C** Could. Items needing a client decision are marked **TBC** .

1.3 Glossary

Term	Meaning
Master data	Reference data: packages, visa types, nationalities
Dynamic field	A user-configurable form field added without code changes
Dynamic print	A printed document driven by an editable template, not hard-coded layout
Voucher	A printed financial document (invoice, receipt, payment)
RBAC	Role-Based Access Control
RTL	Right-to-left text direction (Arabic)
Balance	Net invoice amount – received + refund

2. System Overview & Module Map

The system is organised into transaction modules, master data, supporting engines (dynamic fields, dynamic print, AI), and cross-cutting services (security, audit, reporting). The map below shows how the modules relate.

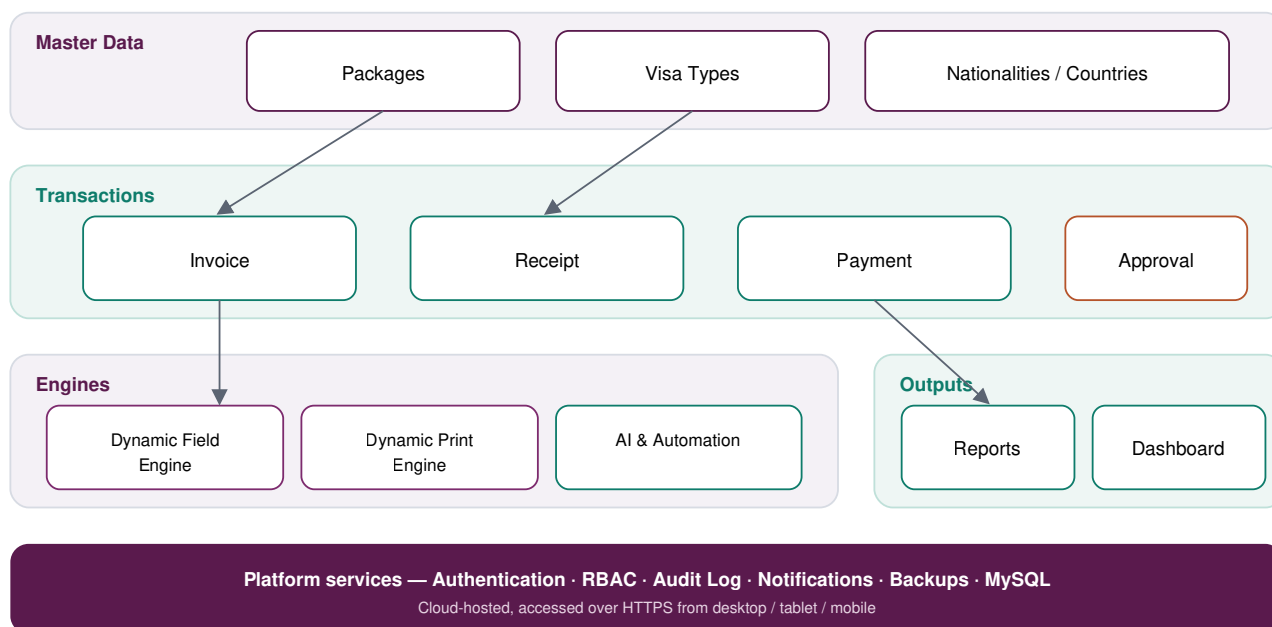


Figure 2.1 — Module map: master data feeds transactions; engines and outputs sit on shared platform services.

2.1 Modules at a glance

Module	Purpose
Master Data	Maintain packages (name + rate), visa types, nationalities — the reference data used everywhere
Invoice	Create the sale: customer, package, passengers, amount, discount, net, status
Receipt	Collect money against an invoice (cash/bank); track balance
Payment	Record office expenses and customer refunds
Approval	Review and approve/reject invoices with comments
Reports & Dashboard	Income, pending, passengers, departure-wise, expense; live KPIs
Dynamic engines	Configure fields and printed documents without code

3. Architecture & Technology

A cloud-hosted web application with a clear separation between the React front-end, a Node.js REST API, and a managed MySQL database. The app tier is stateless and scales horizontally; all access is over HTTPS with token-based authentication and role-based authorisation.

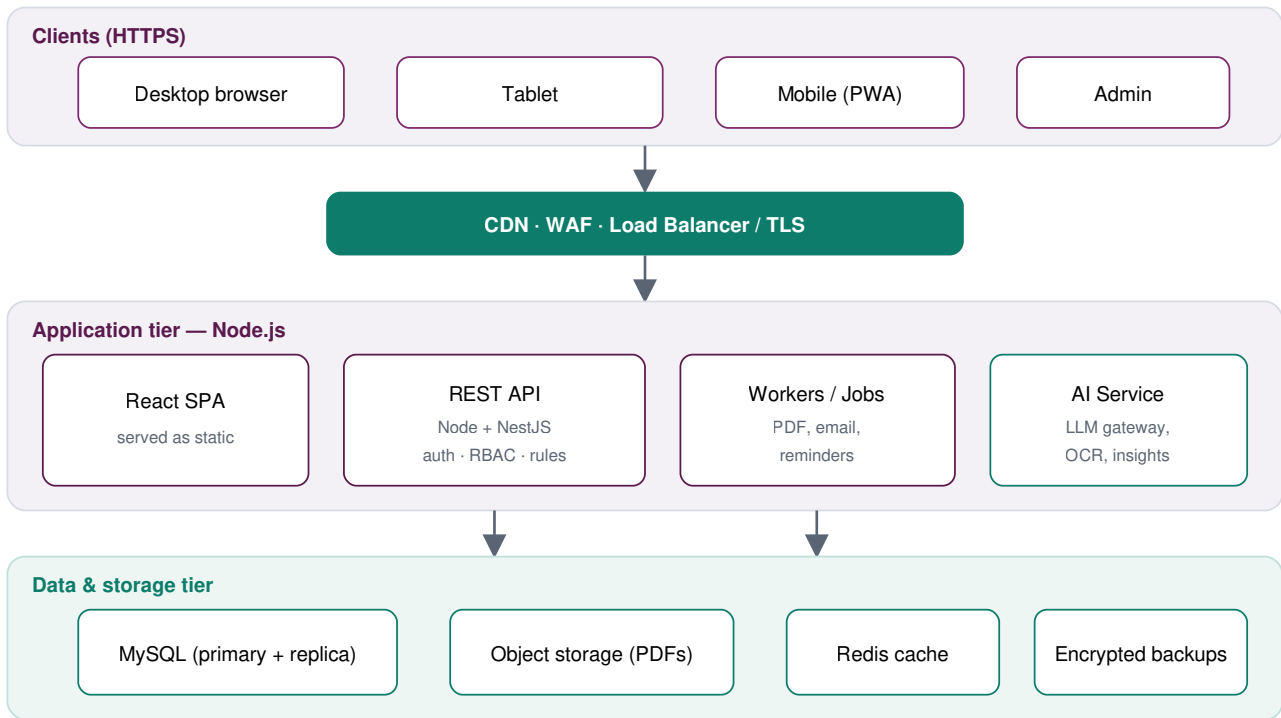


Figure 3.1 — Logical deployment architecture.

3.1 Technology stack

Layer	Technology	Notes
Front-end	React 18 + TypeScript, Vite, i18next (EN/AR + RTL)	Responsive SPA/PWA, bilingual right-to-left support
Back-end	Node.js (LTS) + NestJS (TypeScript)	Structured REST API with validation, guards, modules
Database	MySQL 8 (InnoDB, utf8mb4), Prisma / TypeORM	Transactional integrity; JSON columns for dynamic fields
Cache / queue	Redis + BullMQ	Caching, background jobs (PDF, email, reminders)
PDF / print	Headless Chromium (Puppeteer) + HTML/CSS templates	Pixel-accurate bilingual documents
AI	LLM gateway (provider-agnostic), OCR engine	Document extraction, assistant, insights
Auth	JWT access/refresh, argon2/bcrypt, optional TOTP 2FA	Stateless secure sessions
Infrastructure		

	Docker containers; cloud host/region TBC	Region near Qatar recommended for latency/ residency
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4. Module Specifications

4.1 Master Data — Packages, Visa Types, Nationalities

ID	Requirement	Pri
FR-MD-1	Maintain Packages : name (EN + AR), rate, currency (QAR), status, optional attributes (duration, includes-visa, transport, hotel).	M
FR-MD-2	Maintain Visa Types (e.g. Umrah Visa, Tourist Visa): name EN/AR, status.	M
FR-MD-3	Maintain Nationalities : country name, short code, Arabic name.	M
FR-MD-4	All lists support search, inline filters, sort, pagination, soft-delete (deactivate).	M
FR-MD-5	A master record referenced by transactions cannot be hard-deleted; it is deactivated.	M
FR-MD-6	Bulk import/export of master data via spreadsheet.	S

4.2 Invoice

ID	Requirement	Pri
FR-INV-1	Create/edit invoices with auto number & date, customer, nationality, mobile 1/2, package, departure date, no. of passengers/seats/visas, room type (Normal/Separate/Nil), remarks.	M
FR-INV-2	Capture a passenger grid (SL no, name, visa type, visa-required) with dynamically added rows.	M
FR-INV-3	Compute amount, discount, net amount from package rate × passengers; manual override allowed and audited.	M
FR-INV-4	Track invoice status (Draft / Confirmed / Cancelled) and approval status (Pending / Approved / Rejected) with comments.	M
FR-INV-5	List view with filter/search by no, date, customer, departure, status; multiple invoices open in tabs.	M
FR-INV-6	Print/export invoice to PDF via the dynamic print engine (bilingual).	M
FR-INV-7	Maintain running balance = net – received + refund per invoice/customer.	M

4.3 Receipt / Collection

ID	Requirement	Pri
FR-RCT-1	Create receipts with auto no & date, linked invoice, customer, nationality, mobile 1/2, package, departure, passengers/seats/visas, room details.	M
FR-RCT-2	On selecting an invoice, auto-fill details and show invoice amount, previous balance, remaining balance.	M
FR-RCT-3	Record mode of payment (Cash / Bank) and received amount; prevent over-collection beyond balance unless overridden & audited.	M
FR-RCT-4	Print bilingual Receipt Voucher (سند قبض) matching current format: terms text, received-amount box, package, passenger details, total & remaining balance, three signatures.	M

FR-RCT-5	List, search, reprint; void with reason (audited).	M
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4.4 Payment — Expense & Refund

ID	Requirement	Pri
FR-PAY-1	Record payments of type Expense or Refund with auto no & date, paid-to, mobile, narration, amount, remarks.	M
FR-PAY-2	Refund payments link to an invoice and reduce its income/balance.	M
FR-PAY-3	Optional expense categories (office, salary, transport, tickets) for analysis.	S
FR-PAY-4	Print payment voucher; list, search, edit, delete (audited).	M

4.5 Approval Workflow

ID	Requirement	Pri
FR-APR-1	Invoices carry approval status (Pending → Approved/Rejected) with approver, comment, timestamp.	M
FR-APR-2	Configurable rule: e.g. discounts above a threshold require manager approval. TBC	S
FR-APR-3	Notifications to approvers (in-app / email) and an approvals queue.	S

4.6 Reporting & Dashboard

ID	Report / requirement	Pri
FR-RPT-1	Income Summary — invoice/received/refund/income/balance per invoice with totals.	M
FR-RPT-2	Pending — invoices with outstanding balance.	M
FR-RPT-3	Income Detail — detailed income breakdown.	M
FR-RPT-4	Passengers List — passengers grouped by departure date with invoice & customer.	M
FR-RPT-5	Departure-Date-wise — invoices grouped/filtered by departure with approval status.	M
FR-RPT-6	Expense — payment vouchers with narration and totals.	M
FR-RPT-7	Dashboard — KPI cards (today's collection, pending balance, departures this week, expenses) and charts.	S
FR-RPT-8	All reports date-range driven; export to PDF and Excel/CSV; column show/hide; saved filters.	S

5. Dynamic Field Engine

Fields can be added or changed without a code release. An authorised administrator defines custom fields on any entity (invoice, receipt, payment, customer, passenger, package) through a configuration screen.

5.1 Capabilities

- Field types: text, number, currency, date, dropdown (managed list or master data), checkbox, multi-select, phone, email, file attachment, computed/formula.
- Per field: label (EN + AR), help text, default, placeholder, required, min/max, regex validation, display order/section.
- Role-based visibility and edit-ability; conditional display (“show B when A = X”).
- Mark a field printable and map it into document templates.
- Activate/deactivate or version a field without losing historical data.

5.2 Storage model

Field *definitions* live in metadata tables; field *values* are stored per record in MySQL **JSON** columns, giving flexibility without schema migrations while core columns stay strongly typed for reporting.

Table	Purpose
<code>field_definitions</code>	Entity, key, type, labels (EN/AR), validation, options, role visibility, print flag, order, status
<code><entity>.custom_data</code> (JSON)	Key → value store of custom field values per record
<code>field_audit</code>	Who changed which definition and when

6. Dynamic Print & Document Engine

Every printed document is driven by an editable template, not hard-coded layout. Templates render to pixel-accurate PDF via headless Chromium, fully supporting bilingual English/Arabic content and right-to-left layout.

6.1 Capabilities

- Template editor with company header, logo, address, EN/AR blocks (matching current vouchers).
- Insert any standard or dynamic field as a placeholder, e.g. `{{customer.name}}` , `{{invoice.net_amount}}` , `{{custom.passport_no}}` .
- Repeating sections (passenger lists, line items) and conditional blocks (show terms only on receipts).
- Per-document settings: page size, margins, header/footer, watermark, language(s), signature lines.
- Multiple templates per document type with versioning; set default and preview before publishing.
- One-click reprint, bulk print, email as PDF, download.

6.2 Documents reproduced

Document	Notes
Invoice	Bilingual, passenger grid, amount/discount/net, remarks
Receipt Voucher (سند قبض)	Terms text, received-amount box, package, passenger details, total & remaining balance, three signatures
Payment Voucher	Expense/refund, paid-to, narration, amount
Reports	Income Summary, Departure-wise, Pending, Income Detail, Passengers List, Expense — company header + totals

Bilingual & RTL. The engine handles mixed Latin/Arabic strings, RTL alignment, and Arabic-Indic numerals where required, so new vouchers match the existing ones while remaining admin-editable.

7. AI & Automation

AI is applied where it removes manual effort or reduces error. Features are optional, auditable, and keep a human in control of financial actions. The provider is configurable via an LLM gateway.

Feature	What it does	Pri
Passport / ID OCR	Scan/photograph a passport; auto-extract name, nationality, passport no, expiry; pre-fill the form for staff to confirm.	S
Smart data entry	Suggest packages, detect duplicate customers, validate phone/visa formats, transliterate names EN ↔ AR.	S
Natural-language reports	Ask “how much did we collect this week?” and get an answer plus the underlying report.	C
Collection assistant	Draft polite bilingual payment-reminder messages for outstanding balances.	C
Anomaly hints	Flag unusual discounts, duplicate receipts, out-of-pattern expenses for review.	C
Departure reminders	Notify staff/customers of upcoming departures and missing documents.	S

Guardrails. AI never posts transactions or approves records on its own — it suggests, a user confirms. Personal data sent to AI is minimised; provider/region is a client decision. **TBC**

8. Security & Access Control

8.1 Authentication & access

- Individual accounts, strong password policy, passwords hashed with argon2/bcrypt.
- JWT access/refresh tokens; automatic session expiry/refresh.
- Optional 2FA (TOTP) for admin and finance roles.
- Role-Based Access Control down to module and action level.
- Account lockout after repeated failed logins; forced password reset.

8.2 Roles (initial)

Role	Typical permissions
Administrator	Full access incl. users, roles, dynamic fields, templates, settings
Manager	Approve invoices, view all reports/dashboards, void/refund with reason
Counter staff / Agent	Create invoices, receipts, payments; print; no deletions/approvals
Accountant	Payments, expenses, refunds, financial reports & exports
Viewer / Auditor	Read-only access to records, reports and audit log

8.3 Data protection & auditability

- TLS 1.2+ everywhere (HTTPS only, HSTS); encryption at rest for DB, backups and documents.
- Tenant/branch data isolation; row-level scoping by branch where applicable.
- Daily automated backups with tested restore; point-in-time recovery.
- Full immutable audit trail: who created/edited/deleted/approved what and when.
- Soft-delete with reason for financial records; OWASP Top 10 defences; WAF, rate limiting, dependency scanning.

9. Data Model & Database Design

MySQL schema (InnoDB, utf8mb4 for full Arabic support) with foreign keys and indexed lookups. Core entities below; the dynamic field engine adds metadata tables and JSON value columns alongside.

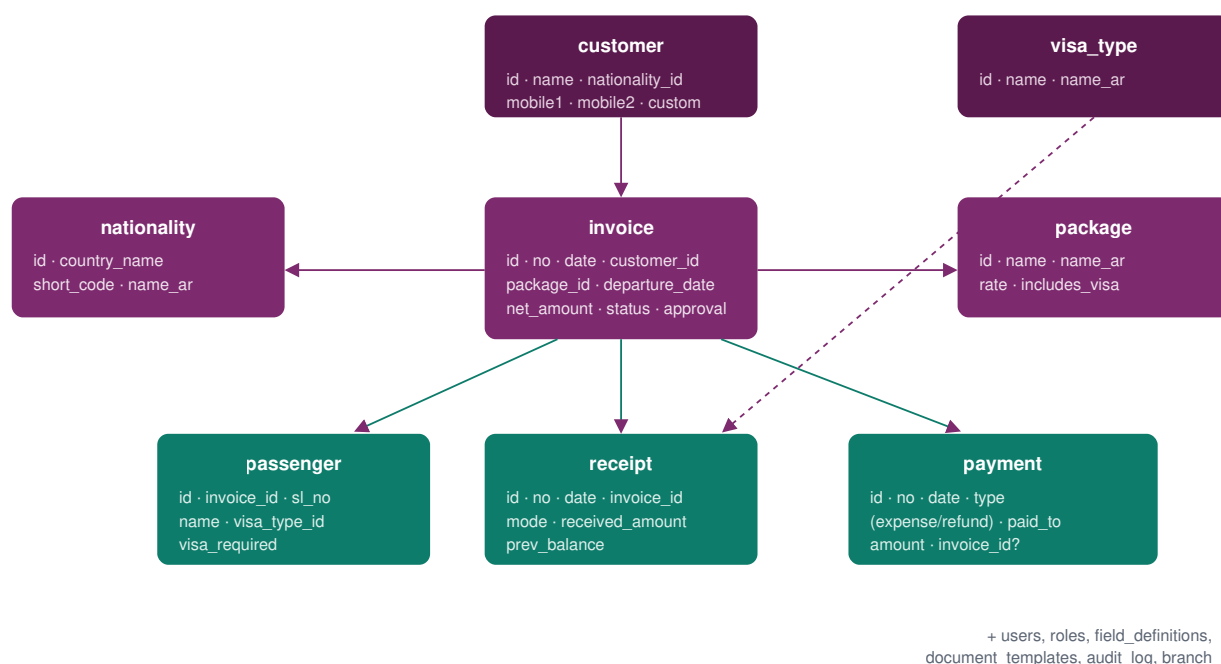


Figure 9.1 — Core entity-relationship overview (simplified).

9.1 Core tables

Table	Key columns
branch	id, name, address, contacts (multi-branch support)
user / role / permission	RBAC: users, roles, permissions, assignments, 2FA
nationality	id, country_name, short_code, name_ar
visa_type	id, name, name_ar, active
package	id, name, name_ar, rate, currency, includes_visa, active
customer	id, name, nationality_id, mobile1, mobile2, custom_data(JSON)
invoice	id, no, date, customer_id, package_id, departure_date, seats, visas, room_type, amount, discount, net_amount, status, approval_status, custom_data
passenger	id, invoice_id, sl_no, name, visa_type_id, visa_required, custom_data
receipt	id, no, date, invoice_id, mode, prev_balance, received_amount, custom_data
payment	id, no, date, type, paid_to, mobile, narration, amount, invoice_id, category_id
field_definition / document_template	Dynamic field & print metadata

audit_log	entity, record_id, action, user_id, before/after, timestamp
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10. Non-Functional Requirements

ID	Requirement
NFR-1	Performance. Common screens load under 2s; report generation under 5s for typical ranges.
NFR-2	Availability. Target 99.5%+ uptime with monitoring and alerting.
NFR-3	Scalability. Horizontal scaling of the app tier; supports growth in branches, users, records.
NFR-4	Usability. Familiar list+detail layout, keyboard-friendly entry, responsive on mobile/tablet.
NFR-5	Localisation. Full EN/AR UI with RTL; Hijri/Gregorian date option. TBC
NFR-6	Compatibility. Latest Chrome, Edge, Safari, Firefox; installable PWA.
NFR-7	Maintainability. Documented API, automated tests, CI/CD, coding standards.
NFR-8	Backup & recovery. Daily backups, tested restore, RPO ≤ 24h, RTO ≤ 4h. TBC
NFR-9	Auditability. All data changes logged and retained for an agreed period.
NFR-10	Accessibility. WCAG-aligned contrast, labels, keyboard navigation.

11. Process Flows

■ Start / actor step ■ System / output ■ Decision

11.1 System context & navigation flow

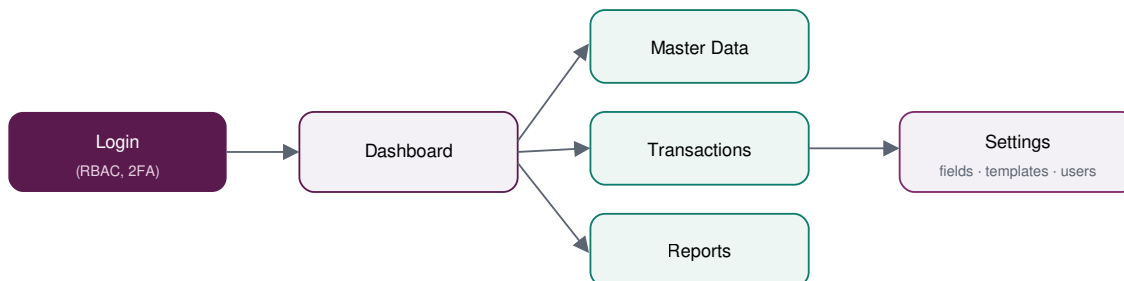


Figure 11.1 — Top-level navigation after login.

11.2 End-to-end operational flow

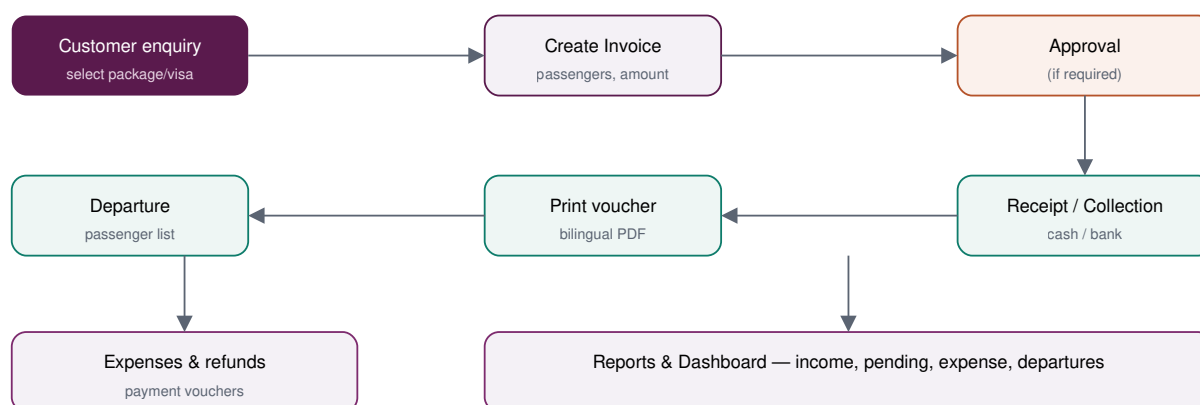


Figure 11.2 — Day-to-day operational flow.

11.3 Invoice creation flow

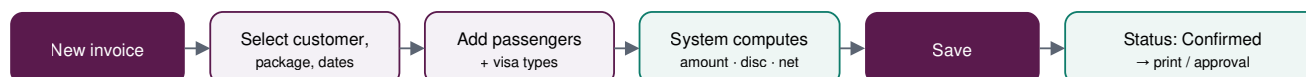


Figure 11.3 — Invoice creation.

11.4 Receipt / collection flow



Figure 11.4 — Receipt / collection against an invoice.

11.5 Payment (expense & refund) flow

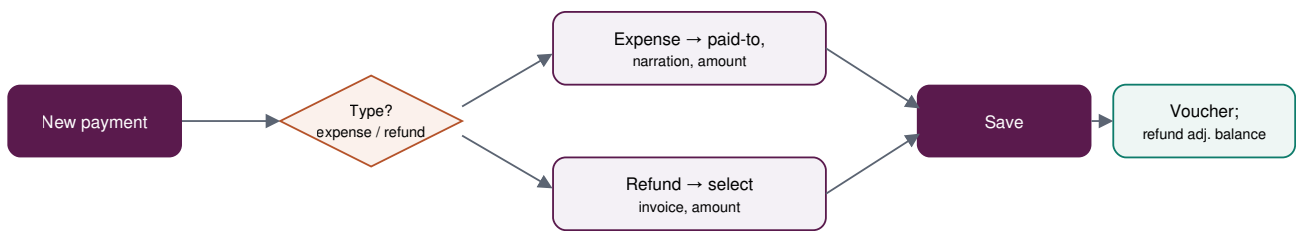


Figure 11.5 — Payment branching into expense or refund.

11.6 Approval flow

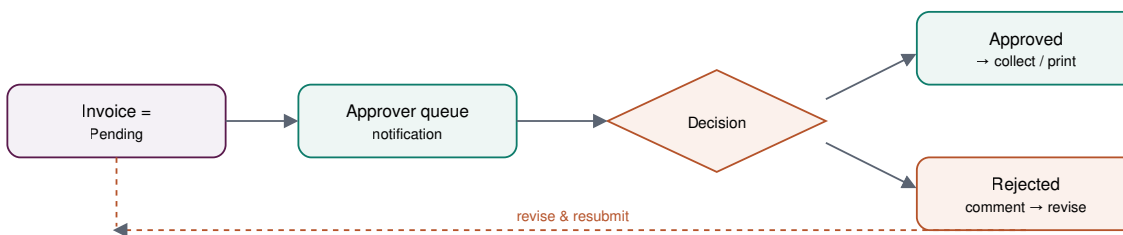


Figure 11.6 — Approval, with reject-and-revise loop.

11.7 Dynamic field & dynamic print flow

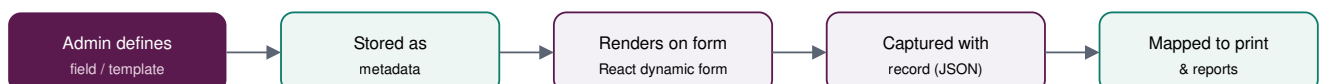


Figure 11.7 — Define-once, use-everywhere flow for dynamic fields and printing.

11.8 Data flow diagram

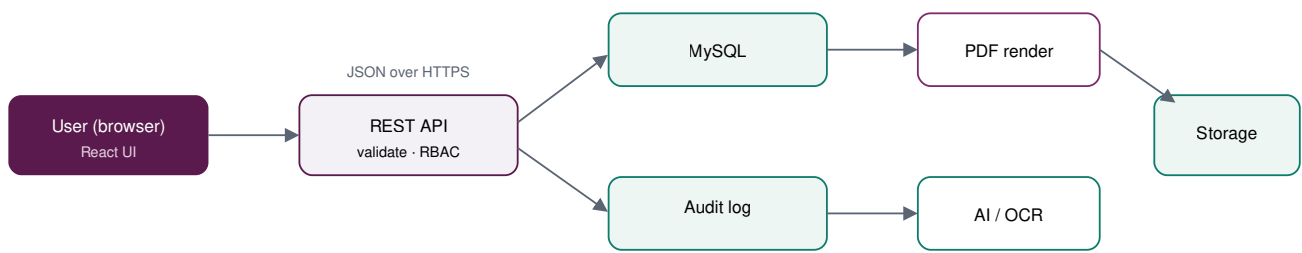


Figure 11.8 — How data moves between UI, API, database, and services.

12. Data Migration

Historical data from the current desktop application is migrated into the new MySQL database so the business keeps a single, continuous record.

1. **Assess source.** Identify the current database engine and export options (direct DB, backup file, or CSV export). **TBC**
2. **Map data.** Map source tables/fields to the new schema (customers, packages, visa types, nationalities, invoices, passengers, receipts, payments).
3. **Build ETL.** Extract, clean (de-duplicate, fix Arabic encoding), transform and load, preserving original document numbers.
4. **Trial migration.** Load into staging; reconcile counts and financial totals against the old system's reports.
5. **Validation.** Verify key records and report totals match.
6. **Cutover.** Final delta migration, freeze old system, go live; keep the old system read-only as a fallback for an agreed period.

Dependency. Migration effort depends on access to the current data. An early database export or sample confirms the mapping.

13. Configuration Parameters (to confirm)

These items are configurable and need a decision before or during build. They do not change the specification, only its settings.

#	Item	Default / note
1	Cloud provider & hosting region (data-residency)	Region near Qatar
2	AI provider and which AI features in phase 1	Provider-agnostic gateway
3	Access to current data for migration	DB export / sample
4	Number of branches and concurrent users	Single branch assumed
5	Hijri date display	Gregorian default
6	Notifications channel (email / SMS / WhatsApp)	Email default
7	Approval rule thresholds (e.g. discount limit)	Off by default
8	Currency & number/date formats	QAR, dd/mm/yyyy

SPEC-RAWDA-ERP-001 • Version 1.0 • Functional & Technical Specification with Process Flows